Nose Wheel-Ski (Mod S.O.O. 6001) – General data

1. General

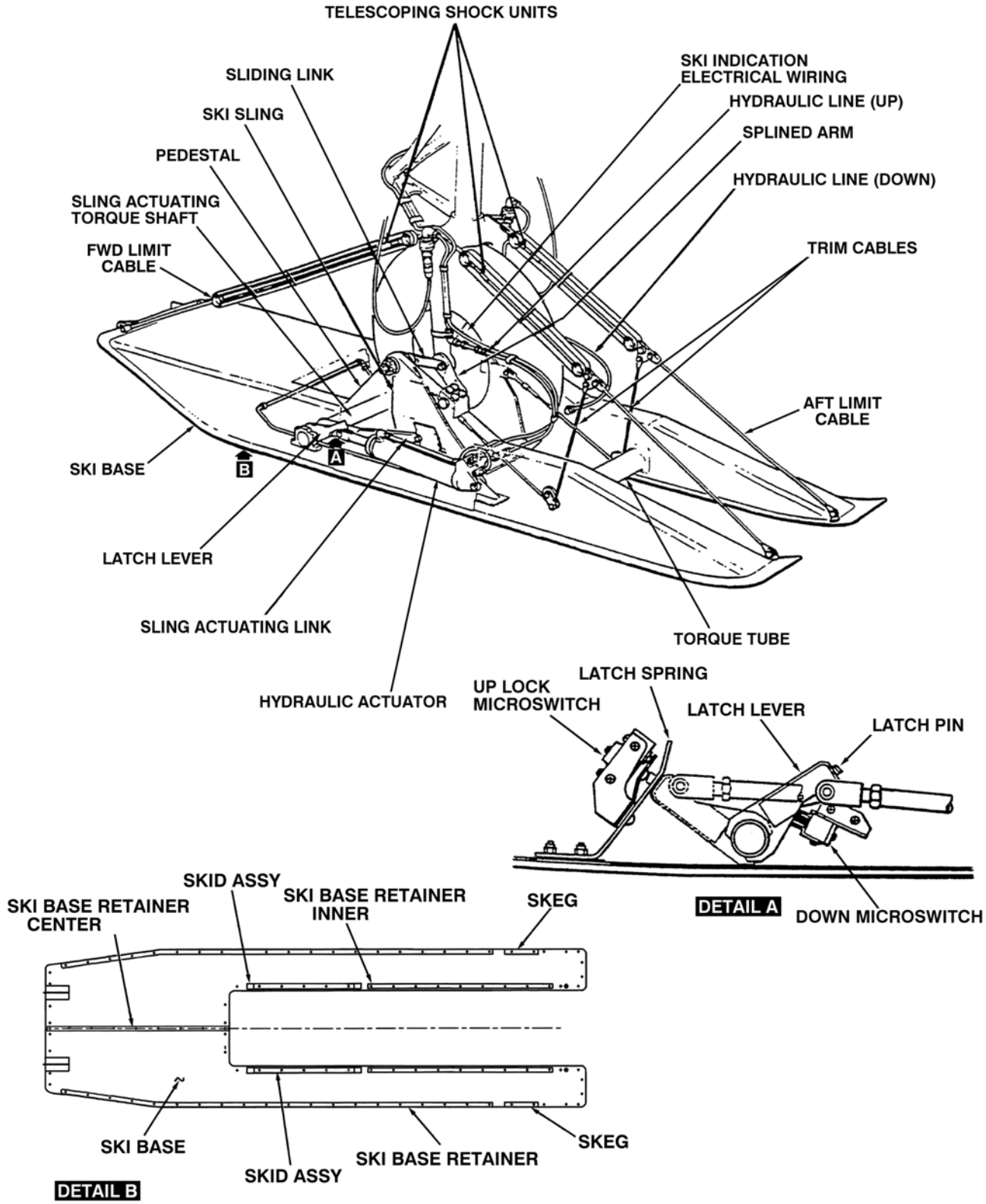
The Nose Wheel-Ski forms part of the Wheel-Ski installation (refer to 32-00-00, Landing Gear – General Data). The ski is attached to the nose wheel axle by two splined arms, and to the nose wheel fork by two pairs of telescoping shock units. The ski, which is controlled by the same lever as the Main Wheel-Skis, is hydraulically actuated from the aircraft's hydraulic system to the “DOWN/SNOW” or “UP/LAND” position (refer to 32-30-00, Extension and Retraction (Wheel-Ski) (Mod S.O.O. 6001) – General data). The ski is rigged level in flight and is self trimming in pitch for either a ski or wheel landing. An indicator, which shows the ski position, is provided on the ski selector panel in the flight compartment. When installing the nose ski the torque loading on the shimmy damper at the steering collar must be increased, and when the skis are removed the torque must be reduced (refer to 32-20-34, Steering Collar – Maintenance Practices or 32-20-21, Steering Collar (with Mods S.O.O. 6001 and 6048) – Maintenance Practices).

The Nose Wheel-ski base, (Mod 6/2199) consists of a UHMW layer retained by a series of steel retainer strips, screws and fittings. The UHMW base is trimmed out around fittings where tension cables are attached. Fasteners run through the retainer strips bolting the base to the ski. The base material provides a low friction surface on the bottom of the ski and helps protect the skis from damage experienced during landings. The retainer strips are wear components and should be inspected in accordance with PSM 1–6–7 and replaced as required.

2. Description

Refer to Figure 1 for details

The Nose Wheel-Ski consists of two beams structurally joined at the front, and connected at the rear by a torque tube secured to the left and right beams. A double-acting hydraulic actuator is housed in the left ski beam, and is connected by its cylinder to the beam structure. The rod end of the actuator connects to a latch lever attached to the end of a sling actuating torque shaft having two arms within the ski tunnel. The two arms are connected to either side of a sling by actuating links. A pedestal on each beam provides the pivot points for the ski sling and forms the mounting for the pivot blocks which attach the ski to the nose gear axle splined arms. Two hydraulic connections are provided, one on the inner side of each beam. The right-hand beam hydraulic pipe is routed through the right beam and the front section of the ski into the left beam, to the “down” side of the hydraulic actuator. The left-hand hydraulic pipe is routed within the left beam to the “up” side of the actuator. Flexible hydraulic hoses are installed between the connections on each side of the ski to pipe connections on the respective sides of the nose wheel shock strut fork. A ski position electrical cable connector is installed on the left ski beam and is wired internally to the two limit microswitches operated by the latch lever. Two trim cables, one connected to each side of the ski sling, are routed over pulleys to their respective rear left and right telescoping shock units. The four telescoping shock units, two connected to eye bolts at the front of the ski and two at the rear, connect to attachment brackets on the nose wheel shock strut fork. Each shock strut unit consists of a telescoping unit provided with positive stops to limit the maximum pitch attitudes of the ski, on which are mounted endless shock cord (bungee) rings, to restrain the ski in the neutral position and to provide a means of retracting the ski when the sling is actuated.



Nose Wheel-Ski
Figure 1

3. Operation

When skis "DOWN/SNOW" is selected, the actuator retracts unlocking the latch and, through the connecting linkage, moves the sling down under the wheel, in doing so the sling impinges on the tire and forces the wheel to roll over the top of it. This action causes the ski to move down into its extended position. Simultaneously as the sling moves rearward, the two trim cables slacken off, thus relieving the tension on the rear trim units. On selecting "UP/LAND", the actuator extends and moves the sling up and forward, allowing the wheel to roll from it. At the same time the trim cables, attached to the sling, tension the rear shock units. When the latch lever reaches the limit of its travel, a plunger on the end of the latch lever engages in a stop plate to lock the ski up. The two pairs of telescoping shock units each serve two purposes. The forward pair, by means of the shock cords, exert a restoring force to oppose external pitch down forces arising from aerodynamic maneuvering or handling loads, and by means of the internal stops, restrict the pitch down angle of the ski. The rear pair, by means of the shock cords, predetermine (after adjustment) the level trim attitude of the ski in fight, and through the trim cables, allow sufficient pitch up travel to accommodate surface irregularities encountered on snow or ice when selected to "DOWN/SNOW". The pitch up angle of the ski is also restricted by the rear shock unit internal stops.

4. Servicing

A. Nose Wheel-Ski Lubrication

Note The lubrication points, method and frequency of application are given in the Nose Wheel-Ski Lubrication diagram, refer to 12-20-10, Lubrication – General Data, Figure 2.

5. Removal/Installation

Refer to Figure 2 for details.

A. Remove Nose Wheel-Ski

Note The following removal procedures are based on the assumption that the aircraft is in the "UP/LAND" configuration.

1) Remove fuselage to landing gear leg fairings (refer to 12-00-00, Servicing – General Data) and, using main inboard jacking points and nose jacking beam (SD12538), jack aircraft until all wheels are at least 8 inches clear of ground (refer to 07-00-00, Lifting and Shoring – General Data).

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.) and ensure HYDRAULIC OIL PUMP circuit breaker is engaged.

Note Observe that motor pump charges accumulators, and that motor cuts out when pressure reaches 1575 ± 50 psi.

3) With the skis selected to the "DOWN/SNOW" position, disconnect sling trim cables from rear telescoping shock unit junction links.

4) Install chocks to forward and rear telescoping shock units.

Note The chocks referred to are supplied with the shock units as an aid to installation. For reference purposes, the length of the chocks are 17 9/16 inches for the rear, and 23 inches for the forward shock units.

5) With the skis selected to the "UP/LAND" position, block up ski level with aircraft datum, and remove rear telescoping shock units.

6) Remove forward telescoping shock units.

WARNING **SERIOUS INJURY COULD OCCUR SHOULD THIS HIGHLY-STRUNG SHOCK UNIT BE ACCIDENTLY DROPPED AND FRACTURED IN AN UNTENSIONED OR UNCHOCKED STATE.**

7) Select POWER SOURCE and MASTER switches OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches OFF (Post Mod 6/2314).

8) Disengage HYDRAULIC OIL PUMP circuit breaker, and discharge hydraulic pressure from accumulators by operating brake pedals.

9) Using a suitable container to collect oil spillage, disconnect and remove hydraulic lines and electrical connections between ski and aircraft. Install blanking caps to open hydraulic lines.

10) Remove pivot bearing blocks attaching splined arms to ski.

11) Remove tire valve access cap and position wheel to provide clearance to remove left center bolt from pedestal pivot. Remove center bolt.

    Note The nose wheel is normally installed with the tire valve on the left, if wheel has been installed with valve on right, for Para 5.A.11) read “right center bolt”, and in Para 5.A.12) read “ski to left” and “left center bolt”.

12) Remove right center bolt nut, move ski to right, and remove right center bolt from pedestal pivot.

13) Remove blocks supporting ski. Remove ski.

14) Remove tie rod and splined arms from axle.

15) Install towing spools into axle and insert axle retaining bolts. Tighten axle retaining nuts and bolts.

16) At steering collar, back off torque on preload nut (refer to 32-20-34, Steering Collar – Maintenance Practices or 32-20-21, Steering Collar (with Mods S.O.O. 6001 and 6048) – Maintenance Practices.

17) If necessary, remove main wheel-skis before carrying out next step (refer to 32-11-00, Main Wheel-Skis (Mod S.O.O. 6001 and Mod 6/2182) – General data).

18) Lower aircraft to ground and remove jacks.

19) Switch on electrical power by selecting POWER SOURCE switch to EXTERNAL or BATTERY and MASTER switch to ON (Pre Mod 6/2314); or by selecting EXTERNAL or BATTERY switch to ON (Post Mod 6/2314). Check that BUS TIE switch is set to NORMAL.

20) Engage HYDRAULIC OIL PUMP circuit breaker.

    Note Observe that motor pump charges accumulators, and that motor cuts out when pressure reaches 1575 ± 50 psi.

21) Using a suitable container to receive oil spillage, bleed the right nose wheel-ski hydraulic line at fuselage, by slackening blanking cap with ski selector lever selected to DOWN/SNOW. When fluid is free of air, tighten blanking cap with fluid still flowing.

22) Repeat Para 5.A.21) on left nose wheel-ski hydraulic line with an UP/LAND selection made.

    Note If main wheel-skis are being removed at the same time as the nose wheel-ski, bleed main wheel-ski system in conjunction with Para 5.A.21) and Para 5.A.22). (Refer to 32-11-00, Main Wheel-Skis (Mod S.O.O. 6001 and Mod 6/2182) – General data).

23) Ensure after bleeding, that blanking caps on hydraulic lines are tight and secure.

24) Replenish hydraulic system as necessary (refer to 12-10-15, Hydraulic System – General Data).

25) Store wheel-skis complete with attachment fittings for future use.

26) Ensure Apex is reprogrammed with standard gear airspeed programming instructions to suit the gear configuration in accordance with 32-10-11, Main Gear Leg – General Data, Para 3.

27) Install taxi light and connect wiring.

B. Remove Nose Wheel-Ski Base (Mod 6/2199)

Refer to Figure 1 for details.

1) Remove the Nose Wheel-Ski from aircraft as per Para 5.A.

2) Remove the screws securing the skids and skegs from bottom of ski.

3) Remove skids and skegs.

4) Remove the screws securing the retainers to ski base.

5) Remove retainers.

Note It is not necessary to remove the Tow Lug Assembly from the forward end of skis.

6) Remove remaining screws securing ski base to ski.

7) Remove ski base.

C. Preparation of Replacement Ski Base Components

1) Clamp together the old ski base to the replacement ski base.

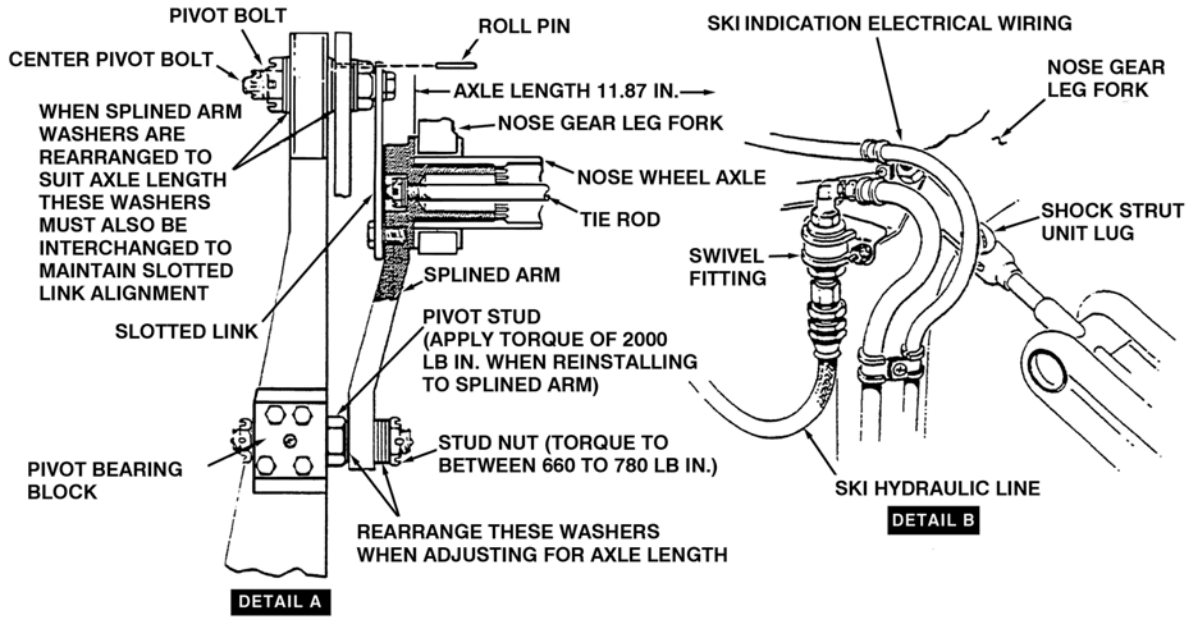
2) Copy profile and hole locations.

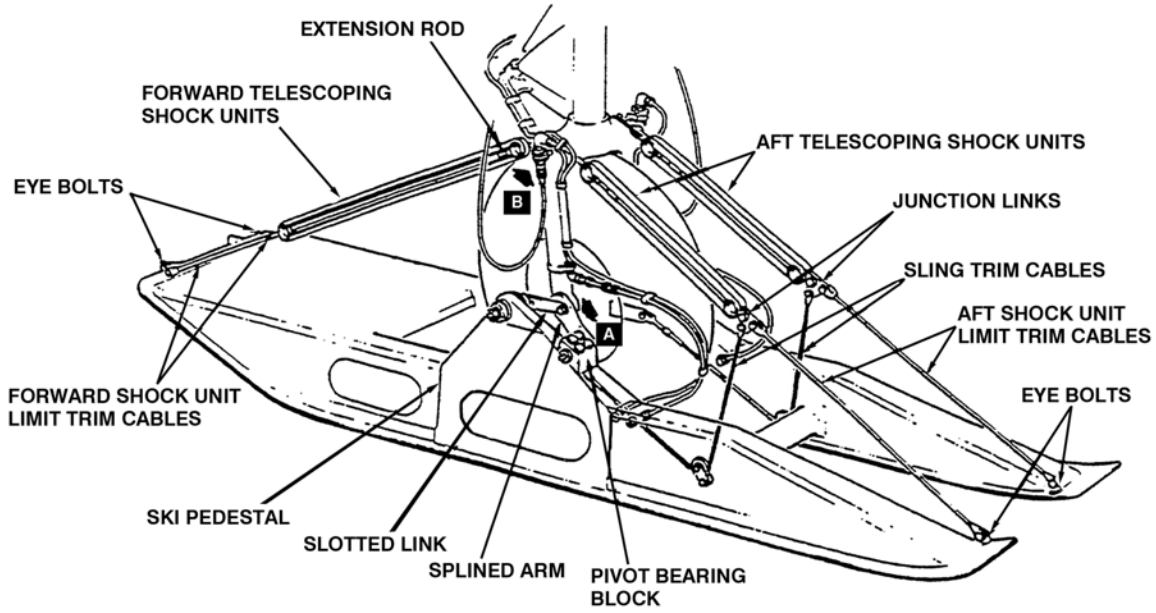
3) Transfer hole pattern from ski to replacement ski base. Match hole size and countersunk as required.

4) Carefully trim replacement base to match old base profile.

5) Using old ski base retainers, bend new retainers to match profile.

6) Clamp retainers securely together and transfer hole patterns. Match hole size and countersink as required.





Nose Wheel-Ski Installation
Figure 2

D. Install Nose Wheel-Ski

Note For the purpose of the following procedures, it is assumed that the aircraft has the necessary structural and nose leg shimmy damper provisions for a wheel-ski installation.

1) Remove fuselage to landing gear leg fairings (refer to 12-00-00, Servicing – General Data) and, using main inboard jacking points and nose jacking beam (SD12538), jack aircraft well clear of ground (refer to 07-00-00, Lifting and Shoring – General Data). Level aircraft (refer to 08-00-00, Leveling and Weighing – General Data).

2) Remove retaining bolts and tow bar spools from nose wheel axle.

3) Check overall length of nose wheel axle. This should agree with the measurement (11.87 inches) between faces of splined arm flanges when splined arms are installed on ski. If axle length is incorrect, obtain correct dimensions by rearranging the washers at pivot stud on either side of splined arm. The washers at the pivot bolts must also be relocated to align and allow freedom of movement of the slotted links. (Refer to Figure 2, Detail A). The pivot stud must be unscrewed from splined arm to relocate washer so when reassembling stud to arm apply a torque of 2000 pound–inches to stud against arm. Torque stud nut to a value of 660 to 780 pound–inches and lock with cotter pin.

4) Inspect entire ski assembly for corrosion and damage and check mechanism for freedom of movement. Lubricate ski (refer to 12-20-10, Lubrication – General Data, Figure 2).

5) Remove both center bolts from the ski pedestal pivots, and four bolts attaching each pivot bearing block to its ski pedestal. Lift off splined arm assemblies complete with slotted links. Separate the two splined arms by removing tie rod.

6) Assemble splined arms (complete with slotted links) to splines inside nose wheel axle, ensuring that arms on each side of wheel are aligned with each other. Check location of tire inflation valve and, because there is insufficient clearance later, insert left or right center pedestal pivot bolt, with washer under head of bolt, through slot in appropriate slotted link.

7) Install tie rod through center of splined arms and axle, tighten tie rod nuts, but ensure tie rod nuts are not torqued to restrict freedom of rotation of splined arms and axle assembly. Secure nuts with cotter pins.

8) With nose wheel-ski in the “UP/LAND” condition (sling up), position ski under wheel. Shift ski sideways as required to facilitate installation of center pivot bolt (refer to Para 5.D.6)), and position wheel tire valve access hole to allow the left center pedestal pivot bolt to be installed.

9) Check before bolting the pivot bearing blocks to ski, that bearing block holes align with holes in ski pedestals. If holes do not align, recheck axle length and distance between faces of splined arms and rectify as in Para 5.D.3).

10) Secure pivot bearing blocks to ski pedestals, with four bolts through each block.

11) Secure left and right pedestal center bolts with nuts and cotter pins. Ensure pivot bolts are not overtightened, slotted links must be allowed to slide freely. Ensure pivot bolts and slotted links are well lubricated.

12) Move ski up and down by hand and check for any indications of binding linkage.

13) Select POWER SOURCE and MASTER switches OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches OFF (Post Mod 6/2314).

14) Disengage HYDRAULIC OIL PUMP circuit breaker and release hydraulic accumulator pressure by operating brake pedals.

15) Install left and right hydraulic pipes, and ski indication electrical wiring to nose gear leg fork. Install telescopic shock strut lugs on fork shoulder (refer to Figure 2, Detail B).

16) Connect left and right hydraulic pipes to ski and fuselage connections.

17) Connect ski indication electrical wiring to ski and nose gear leg connections.

18) Bleed nose wheel-ski hydraulic system as follows:

    a) Switch on electrical power by selecting POWER SOURCE switch to EXTERNAL or BATTERY and MASTER switch to ON (Pre Mod 6/2314); or by selecting EXTERNAL or BATTERY switch to ON (Post Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

    b) Engage HYDRAULIC OIL PUMP circuit breaker.

        Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi.

    c) Select ski selector to DOWN/SNOW, and using a suitable container, bleed the right nose wheel-ski hydraulic line by slackening flexible hose between ski and fork. When fluid is free of air retighten hose while fluid is still flowing.

    d) Repeat Para 5.D.18).c) on left hose making an UP/LAND selection.

        Note If main wheel-skis are being installed at the same time as the nose wheel-ski, bleed main wheel-skis in conjunction with Para 5.D.18) (refer to 32-11-00, Main Wheel-Skis (Mod S.O.O. 6001) – General Data).

19) Hydraulically actuate skis to the “DOWN/SNOW” position (sling down).

20) Position and firmly block up nose wheel-ski until it is parallel to the aircraft datum and sling is in firm contact with tire.

21) On left and right sides of nose wheel-ski measure distance between centers of eye bolt hole at aft end of ski and hole in rear lug on nose gear leg fork shoulder.

22) With chocks fitted, preassemble left and right aft telescoping shock units and cables with junction links.

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23) Adjust fork at upper end of each aft telescoping shock unit, until overall length of each assembly agrees with measurements taken in Para 5.D.21), taking into account that assembly will shorten approximately 1/16 inch, when the wood chocks are removed.

24) Install telescoping shock unit assemblies between nose gear leg fork lugs and eye bolts at rear of each ski beam. Do not remove chocks.

25) Actuate ski to the “UP/LAND” position and firmly block up ski level with aircraft datum +2° –0°. Center pivot bolts must contact end of slot in slotted links.

26) On left and right sides of nose wheel-ski, measure distance between centers of eye bolt at front of ski and forward lug attachment on nose gear leg fork shoulder.

27) With chocks fitted preassemble left and right forward telescoping shock units and forward limit cables.

28) Adjust fork at upper end of each forward shock unit assembly until overall length agrees with dimension obtained in Para 5.D.26).

29) Install telescoping shock unit assemblies between forward nose gear leg fork lugs and eye bolts at front of ski.

30) Remove wood chocks from all four telescoping shock unit assemblies.

31) Actuate ski to the “DOWN/SNOW” position and mark the forward shock unit extension rods. Tilt ski nose down to ensure the minimum overtravel of 1/2”. If required, readjust length of forward telescoping shock unit assemblies.

    Note It is necessary to replace chocks and return skis to the “UP/LAND” position, in order to carry out adjustment in Para 5.D.31).

32) Recheck rigging, making adjustments as necessary to the aft telescoping shock units, to maintain skis level with aircraft datum +2° –0°, ensuring that they remain under equal tension and that neither is pulled from its minimum limit length.

    Note These units are normally non-performing, but provide the necessary additional travel (2 inches) to accommodate severely undulating motions sometimes encountered in the “ski - landing” and “ski - taxiing” cases.

33) Loosely connect the sling trim cables between the sling attachments and the telescoping shock strut assembly junction links, through pulleys.

34) Actuate ski to the “wheel - landing” position and, with hydraulic pressure maintained, adjust both sling trim cable turnbuckles to position ski level with the aircraft datum +2° –0° ensuring equal tension of both cables.

    Note Turnbuckle adjustment can be facilitated by pitching nose of ski down.

35) Functionally test main and nose wheel-skis (refer to 32-30-00, Extension and Retraction (Wheel-Ski) (Mod S.O.O. 6001) – General data).

    Note In the “ski - landing” position the ski will assume an attitude approximately 2° nose up.

36) Lockwire the forks at the upper ends of the shock units so as to prevent chafing or fouling the bungee rings.

37) Test wheel-ski indication system (refer to 32-61-00, Wheel-Ski Indication (Mod S.O.O. 6001) – General data).

38) At steering collar, increase torque loading at preload nut (refer to 32-20-34, Steering Collar – Maintenance Practices or 32-20-21, Steering Collar (with Mods S.O.O. 6001 and 6048 – Maintenance Practices).

39) With skis selected to the desired position lower aircraft to ground, remove jacking and leveling equipment. Install fuselage to leg fairings.

40) Check hydraulic system fluid level (refer to 12-10-15, Hydraulic System – General Data).

41) Ensure Apex is reprogrammed with “Wheel-Ski” airspeed programming instructions to suit the gear configuration in accordance with 32-10-11, Main Gear Leg – General Data, Para 3.

E. Install Nose Wheel-Ski Base (Mod 6/2199)

1) Ensure bottom of ski is clean and undamaged.

2) Install base to ski and secure with new hardware in countersunk holes on base.

    Note Over torquing of screws will damage holes in UHMW ski base.

3) Install center ski base retainer. Start at forward end of ski and use new hardware.

4) Install the inner ski base retainers. Start at forward end of ski and use new hardware.

5) Install the outer edge ski base retainers. Start at forward end of ski and use new hardware.

6) Install skids and skegs with associated hardware.

6. Adjustment/Test

Note The access panels form part of the stressed structure of the ski and must be reinstalled as soon as adjustments are completed.

A. Adjust Sling Actuating Mechanism

1) Jack up aircraft (refer to 07-00-00, Lifting and Shoring – General Data).

2) Remove forward left-hand panel.

3) Select ski “UP/LAND” position and observe behavior of ski and latch mechanism.

4) If latch pin does not engage locking hole of latch spring, disconnect and adjust fork end of hydraulic actuator until latch pin engages locking hole.

5) If latch pin engages locking hole of latch spring, release hydraulic pressure and check that clearance between tire and sling is between 0.10 inch and 0.60 inch.

Note Tire clearance is adjusted during manufacture and assembly, and should not be affected appreciably in normal service or by the replacement of related components. Ensure that inadequate tire clearance has not been caused by damage to the latch mechanism or ski bottom in area of latch assembly before attempting tire clearance adjustment.

6) Adjust tire clearance, if necessary, by removing or adding shims under base of latch spring (removing shims increases tire clearance).

Note Adjusting fork end of hydraulic actuator will not alter tire clearance and will only lead to erratic or faulty latch function. If, however, shims have been added or removed, adjustment of hydraulic actuator is also required as in Para 6.A.7).

7) Adjust fork end of hydraulic actuator to give 0.015 inch to 0.060 inch overtravel after latch engagement.

8) Check adjustment of Ski Up and Ski Down limit switches. (Refer to 32-61-00, Wheel-Ski Indication – General Data).

9) Reinstall access panel.

10) Lower aircraft.

7. Inspection/Check

A. Check Nose Wheel-Ski Rigging

1) Remove fuselage to leg fairings (refer to 12-00-00, Servicing – General Data), jack aircraft well clear of ground using inboard jacking points and nose jacking beam SD12538 (refer to 07-00-00, Lifting and Shoring – General Data). Level aircraft (refer to 08-00-00, Leveling and Weighing – General Data).

2) Switch on electrical power by selecting POWER SOURCE switch to EXTERNAL or BATTERY, and MASTER switch to ON (Pre Mod 6/2314); or by selecting EXTERNAL or BATTERY switch to ON (Post Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

3) Ensure GEAR IND and HYDRAULIC OIL PUMP circuit breakers are engaged.

Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi.

4) Select ski selector lever to “UP/LAND”. Check ski is level with aircraft datum +2° –0°, and that center bolts at sling pivot are at end of slot in slotted link.

5) Actuate ski to the “DOWN/SNOW” position and mark the forward shock unit extension rods. Tilt ski nose down to ensure the minimum overtravel of 1/2”. If required, readjust length of forward telescoping shock unit assemblies.

   Note If an adjustment is required to achieve the 1/2 inch overtravel, adjust in accordance with ski installation procedures (refer to Para 5.D).

6) Check that neither of the aft telescoping shock units are pulled from their minimum length.

7) Select ski selector lever to UP/LAND and check that sling trim cables position ski level with aircraft datum +2° –0°.

8) At steering collar check, by applying spring balance to end of torque link, with steering actuator disconnected, that torque required to turn steering collar is between 800 and 1000 pound–inches or 70 to 90 pounds at end of torque link. Reconnect steering actuator.

9) Lower aircraft to ground, remove jacking equipment. Install fuselage-to-leg fairings.

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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